SCI-NOI v0.1

SCI-NOI v0.1 Stage B Shared Normative Core

Implementation-blind Stage B draft

SCI-NOI_NORMATIVE_CORE v0.1/draft-r0.1

Not owner-accepted or frozen. No implementation conformity, validation, calibration, physical-noise, covariance-completeness, significance, performance, readiness, or production claim.

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Normative terms: shall is required, shall not is prohibited, may is permitted only inside the stated boundary, and unavailable is an explicit scientific state rather than a numerical sentinel.

A. Scope, identity, and ownership

NOI-REQ-001 SCI-NOI shall expose NOI-GEN, NOI-UNC, and NOI-STD as separate scientific operations. No operation shall automatically realize, authorize, or mutate another.

NOI-REQ-002 The ordinary GEN method shall be NOI-GEN/PTC-TO-FROZEN-MAP-CONDITIONAL-SIGN@1, with the exact retained PTC occurrence as earliest immutable parent, the exact PTC-to-MAP numerical boundary as insertion point, and the complete frozen MAP accumulation as host. The output shall be an NOI realization map, not an ordinary MAP science product.

NOI-REQ-003 Every GEN method shall classify every scientifically consequential adjacent state as immutable/fixed, rerun/relearned, not applicable, or unavailable. Fixed and relearned members, or members from different replay graphs, shall not share an ensemble.

B. Ordinary finite assignment design

NOI-REQ-004 One ordinary coherence unit shall be one stable realized detector/channel within one exact observation. Its sign shall apply to every admitted PTC occurrence of that detector in that observation. Canonical order shall be lexicographic by canonical observation UID and stable realized detector/channel UID.

NOI-REQ-005 For every observation/readout-network stratum h, the design shall derive each detector mass B_d=sum_p sum_{i in C_p,detector(i)=d} G_pi gamma_i from the exact frozen MAP-admitted positive contribution population. One stratum shall not balance another. B_d shall not be relabeled as precision, empirical uncertainty weight, exposure, support, validity, or a PTC/MAP coefficient.

NOI-REQ-006 Each stratum shall bind an explicit exact rational tolerance tau_h satisfying 0 <= tau_h < 1, with no default. A candidate s_h shall be admissible exactly when abs(sum_d s_d B_d) <= tau_h sum_d B_d. The comparison shall use exact rational arithmetic over the persisted numerical representations of the positive contributions and tolerance; traversal-dependent floating summation shall not decide admission.

NOI-REQ-007 The target one-member law shall be uniform over every admissible sign vector in each stratum and the product of those laws across strata and observations. Candidate signs before conditioning shall be independent uniform values in {-1,+1}. The admissible law shall remain complement-symmetric; it shall not imply detector independence or detector-count balance.

NOI-REQ-008 Resolved members shall be sampled with replacement from the target law. Cross-member streams and observation/network strata shall use disjoint canonical key domains. Complement pairing shall not be forced. Duplicates and complements shall not be silently rejected, replaced, or collapsed.

NOI-REQ-009 Every plan shall explicitly bind a content-identified random-bit generator and algorithm version, opaque seed/key bytes and namespace, B_requested, and a positive per-member/per-stratum retry cap. No generator, seed, count, cap, or tolerance shall be inferred. Missing or conflicting plan facts shall make design resolution unavailable.

NOI-REQ-010 Canonical assignment-key serialization shall use NOI-KEY-C14N@1: type-tagged, length-prefixed UTF-8 NFC fields in this order: SCI-NOI version; route method/version; design generation; earliest immutable parent product/application generation; observation UID; stable array/group; stratum type and owner identity; stratum identity; stable unit identity; member identity; candidate counter; seed/key namespace; random-bit generator and version. Integers shall be canonical base-10 without leading zeros except 0; Boolean and unavailable tokens shall be fixed lowercase; no locale, platform path, whitespace normalization, container index, floating display, or implicit absent value shall participate.

NOI-REQ-011 The design shall use bounded conditional rejection. Candidate rejections before admission shall be construction outcomes, not members or failures. If any required stratum has no admissible assignment or no candidate is accepted within its retry cap, the complete design resolution shall fail closed without tolerance relaxation, cross-stratum balancing, fallback, or a partial design.

NOI-REQ-012 Assignment equality shall require byte-identical canonical sign serialization on the identical ordered coherence domain and design generation. Duplicate detection shall compare a SHA-256 digest first and then require byte equality. Complements shall be paired-but-distinct for full-distribution identity. The product shall report B_requested, B_resolved, B_completed, B_unique, complement-orbit count, B_admitted_for_UNC, design rank, and use-specific effective information separately.

NOI-REQ-013 For the initial method, design rank shall be the exact real rank of the uncentered admitted member-by-coherence sign matrix on its canonical domain. The product shall report its null space or an exact basis/identifier. The known-zero second-moment method shall require at least one admitted member and rank at least one; estimator uncertainty shall be reported or explicitly unavailable. Count shall not substitute for rank or independence.

NOI-REQ-014 A successfully resolved design shall set B_resolved=B_requested>0 and assign every resolved member the exact weight omega_b=1/B_resolved. Failed design resolution shall publish no member weights. A later design with unequal weights shall require a new method version.

C. GEN product and lifecycle

NOI-REQ-015 The exact scientist-readable product class shall be source-bearing conditional randomization ensemble. Its declaration shall record parent source content, suppression target and assumptions, finite balance residual, support/coefficient/projection/filter variation, structured residuals, source-model use/error, leakage, and claim limits. It shall not be called source-free, a repeated physical-noise ensemble, a calibrated null, variance, covariance, precision, or significance by existence.

NOI-REQ-016 Every admitted assignment shall complete through the declared frozen operator. Any incomplete, failed, or unavailable admitted member shall fail the whole ensemble for all UNC use. Completed survivors and partial streaming accumulators shall carry no UNC authority. A retry or replacement shall use a new exact design/generation identity.

NOI-REQ-017 The plan shall select exactly one persistence mode: persisted ensemble, compact deterministic regeneration, or streaming sufficient statistics. Requested, effective, applied, and realized mode shall be distinct; no default or silent fallback is permitted. Each mode shall retain the exact identity, completion, limitations, and sufficiency facts required by its published products and claims.

NOI-REQ-018 An externally owned deterministic transformation shall be used only when its scientific owner supplies exact content-bound purpose, operator, state, parameters, order, domain, support/edge/missing-data behavior, normalization, units, response, validity, lifecycle, and failure authority. NOI shall apply it identically to every compatible admitted realization and shall not choose, tune, substitute, relocate, simplify, or silently omit it.

NOI-REQ-019 A Wiener or FRUIT transformation frozen before member application shall remain a fixed owner-transformed route. Use of an NOI product to learn, select, continue, or update the owner process shall create new immutable transformation, science-product, GEN, and UNC generations. Per-member learning or replay shall be a distinct relearned method. Fixed and relearned members shall not mix.

D. UNC target, estimator, representation, and inverse

NOI-REQ-020 Every UNC method shall bind its exact target law, admitted GEN method/generation and complete membership, center, estimator, design normalization, missingness/dependence treatment, domain, WCS, support, response, unit/beam, representation, rank/null space, regularization, inverse domain, calibration/omissions, estimator uncertainty, lifecycle, claim, and named use.

NOI-REQ-021 UNC shall consume only an all-members-successful GEN ensemble admitted by both the exact member and ensemble policies for the named method. Rejected candidates, failed ensembles, survivor subsets, mixed methods, partial persistence, and partial streaming state shall not be admitted.

NOI-REQ-022 The initial UNC method shall compute, on its exact admitted domain, V_hat_cond(p)=sum_b omega_b M_b(p)^2 with the design weights required above. Its center shall be known zero; it shall not subtract the finite ensemble mean or apply B-1.

NOI-REQ-023 The initial estimator domain shall be D_common={p: every admitted M_b supplies a valid finite value at p}. Outside D_common it shall be unavailable. It shall not use a smaller member subset, pairwise population, zero fill, interpolation, or domain extension.

NOI-REQ-024 V_hat_cond shall have squared signal units and shall be labeled a conditional randomization second moment retaining source imprint and structured residual content. It shall report dependence, complements, all counts, exact design rank, use-specific effective information, and estimator uncertainty or its unavailable state. It shall not be promoted to physical- noise variance, MAP covariance, precision, or significance.

NOI-REQ-025 Retained ensemble, fixed projection, marginal variance, stationary/kernel, block, spectral, sparse, low-rank, full covariance, and unavailable representations shall be separately versioned methods. Every covariance method shall declare estimator, common member population, domain, support, response, rank/null modes, regularization/approximation, omissions, and uncertainty/calibration. Unreported covariance shall remain unknown or unavailable, never zero or independence.

NOI-REQ-026 The only initial inverse product shall be NOI-UNC/INVERSE-CONDITIONAL-SECOND-MOMENT-SCALE, with W_hat_cond=1/V_hat_cond on the exact finite strictly positive parent domain. It shall have inverse squared signal units and shall be unavailable for zero, negative, nonfinite, unavailable, or outside-domain input. It shall not be inverse variance, precision, validity, support, exposure, a PTC/MAP coefficient, or an implicitly regularized value.

E. STD product

NOI-REQ-027 The initial STD method shall be NOI-STD/MAP-CONDITIONAL-SECOND-MOMENT-SCALE@1. It shall bind the exact immutable normalized real-observation MAP signal from the same frozen operator state and compute sigma_cond=sqrt(V_hat_cond) and S_cond=q_MAP/sigma_cond only on the exact compatible finite-positive valid intersection. No interpolation, substitution, implicit inverse route, or regularization is permitted.

NOI-REQ-028 The STD output shall be empirical_scale_standardized_signal with unit exactly 1, explicit numerator/scale dependence, and the sole claim "MAP signal standardized by the stated conditional randomization second-moment scale." It shall not claim a Gaussian, Student, z, N-sigma, probability, detection, completeness, purity, catalog, uncertainty, or JINC meaning.

F. Use-specific admission profiles

NOI-REQ-029 Every NOI-owned admission evaluation shall retain separately named request, applicability, eligibility, and realization fields. Only requested + applicable + eligible + realized shall project to the exact profile action. A generic flag or another-use pass shall have no universal veto, rescue, or propagation effect.

NOI-REQ-030 SCI-NOI:generation_input_admission@1 shall permit only the exact candidate occurrence for the selected PTC-to-frozen-MAP GEN method. Assignment, host application, member completion, ensemble completion, and UNC admission shall remain separate.

NOI-REQ-031 SCI-NOI:uncertainty_member_admission@1 shall admit one exact GEN member only as a candidate for one named UNC method. It shall consume but shall not redefine GEN completion, failure, duplicate/equivalence, support, source-imprint, QC, persistence/reconstruction, lifecycle, cause, or provenance facts. Ensemble admission shall remain separate.

NOI-REQ-032 SCI-NOI:uncertainty_ensemble_admission@1 shall admit only one exact complete all-members-successful ensemble to one exact UNC estimator and domain. For the initial method it shall authorize only the zero-centered design-weighted second moment on the common all-member domain, and for the initial inverse only the exact finite-positive reciprocal domain.

NOI-REQ-033 SCI-NOI:standardization_admission@1 shall permit only construction of the exact initial MAP S_cond product on its compatible finite-positive intersection with the claim in NOI-REQ-028.

G. Unavailability, immutability, and claim ceiling

NOI-REQ-034 Missing, conflicting, incomplete, unsupported, invalid, or failed required facts shall produce the typed state applicable to their scope. No undocumented zero, NaN, infinity, empty object, filename, equal shape, approximate WCS, or hidden default shall create a successful identity join.

NOI-REQ-035 PTC, MAP, JINC, external transformation, Wiener, and FRUIT parents shall remain immutable. Every NOI or successor result shall be a new versioned companion with exact parent, method, generation, dependence, and lifecycle identity.

NOI-REQ-036 No NOI assignment, second moment, covariance, inverse, precision, scale, or weight shall become validity, support, exposure, or a PTC/MAP analysis, gridding, or coadd coefficient without separately authorized successor or feedback authority.

NOI-REQ-037 This draft shall establish no implementation conformity, representation fidelity, numerical validation, calibration, physical-noise validity, covariance completeness, Gaussian significance, achieved performance, readiness, freeze, production suitability, or production authorization. Every unavailable numerical route shall remain unavailable.